

Hyperloop Development Program

Reporting Period 5: Jan 2023 – Jul 2023

31-08-2023

V1.0

This report highlights the progress of the Hyperloop Development Program. First section summarizes the highlights within the growing hyperloop market, the second section provides an overall program overview, followed by section of progress per working group, and summarized in the financial overview.

REPORTING ON HIGHLIGHTS WITHIN HYPERLOOP MARKET

The past months have shown highlights in the Global developments in the Hyperloop Market:

- Jan:
 - o CASIC conducts 50+ km/h hyperloop test in Datong, Shaanxi Province in 210m long tunnel
 - o TU Delft, TU Eindhoven, ETH Zürich and EuroTube start collaboration on mobile factory to build hyperloop tubes
 - o Mercon starts manufacturing and assembling hyperloop pipes for European Hyperloop Center
- Feb:
 - o HyperloopTT SPAC to become a publicly traded company is cancelled
 - o CNN covers hyperloop as how we'll travel in 2050, featuring Hardt Hyperloop
 - o Hardt Hyperloop tests new propulsion prototype for hyperloop vehicle
- Mar:
 - o TUM builds 24-meter hyperloop demonstrator
 - o Hardt Hyperloop signs a strategic collaboration agreement with Posco International
 - o TUM Hyperloop finishes the build of a passenger-size hyperloop demonstrator
- Apr:
 - o Hyperloop appears in Star Wars episode - The Mandalorian — "Chapter 22: Guns for Hire"
 - o China selects Shanghai – Hangzhou as first hyperloop line
 - o Municipal council of Rotterdam accepts motion to study a hyperloop pilot in Rotterdam
 - o Hardt and TuTr sign MoU to converge technologies and collaborate on development roadmap
 - o Berenschot completes second exploratory study about hyperloop with focus on development paths in The Netherlands
- May:
 - o Indian Railways launches €1M tender for consultant to support development of €50M test track in India
 - o Castilla-La Mancha selected as site for €50M test track in Spain
 - o Webuild consortium wins CAV tender for €4M study for €800M hyperloop test track in Veneto in Italy
 - o Globo, the largest commercial TV network in Latin America, features hyperloop as 'train of the future' in primetime show Fantástico
- Jun:
 - o Independent Hyperloop study published by Roland Berger
 - o The Hyperloop Conference was held in Korea

REPORTING ON OVERALL PROGRAM PROGRESS

This report is to confirm costs incurred and progress in the Hyperloop Development Program over the period Jan 2023 – Jul 2023.

The key program deliverable and key deliverable for this stage per working group are represented below.

WG	Key Program deliverable	Key deliverable Status
WG Cargo	Demonstrated hyperloop for cargo system (TRL-7) in the EHC	Design finished, started vehicle manufacturing
WG EHC	European Hyperloop Center (EHC) Phase A completed	Started on-site construction in Veendam
WG FP	Identify and develop 5 routes up to a preliminary level, of which 1 with a go ahead build-out of route	All Preliminary route studies concluded, not yet go-ahead for a pilot route
WG Passengers	Passenger hyperloop system concept (TRL-3 & SRL-5)	System Concept Design Definition (on-hold for now)
WG Program Management	Establish a connected hyperloop ecosystem	Work in progress

HDP is continuing Stage 3 of a total of four stages – as there have been encountered delays, the stage 3 is extended till Jul 2024. The progress of the Working Groups and Work Packages is presented in figure 1. Green means the Work Package is delivered; yellow the Work Package is on-going.

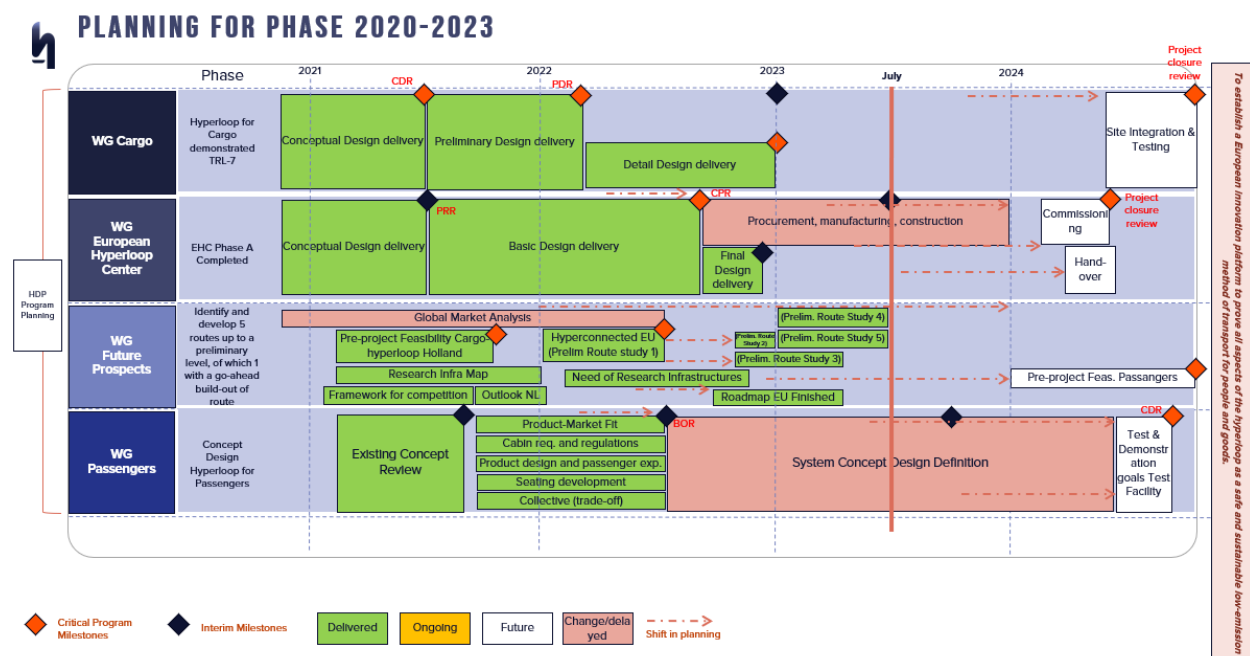


Figure 1: Stages & Milestone planning HDP

The detailed overview of program deliverable status for open stage 1, stage 2, and stage 3 is presented below. A number of milestones from stage 1 and 2 have encountered delays, all adjusted delivery dates are presented below.

#	Milestone	Working Group	Status	(Adj.) Date delivery
Open from Stage 1:				
1.9	Final Market opportunity attractiveness	Future Prospects	Ongoing	Q4-2023
Open from Stage 2:				
2.4	Partners/suppliers contracted for Manufacturing & Assembly.	EHC	On-going (Delayed)	Q4 2023
2.8	Scheme of operations for Hyperloop Research Infrastructure	Future Prospects	On-going (Delayed)	Q4 2023
Stage 3:				
3.5	Preliminary Route #2 (Hyperconnected Europe)	Future Prospects	Finished	Q4 2022
3.6	Preliminary Route #3 (Delhi – Ahmedabad)	Future Prospects	Finished	Q1 2023
3.7	Preliminary Route #4 (Italy)	Future Prospects	Finished	Q2 2023
3.8	Preliminary Route #5 (The Netherlands)	Future Prospects	Finished	Q2 2023
3.9	EHC Delivery & Start first tests	EHC	Not Started (Delayed)	Q1/Q2 2024
3.10	Project hand-over EHC	EHC	Not Started (Delayed)	Q1/Q2 2024
3.11	Roadmap EU Finished	Future Prospects	Delivered	Q1 2023
3.12	Work Package Research Infra finished	Future Prospects	Not Started (Delayed)	Q4 2023

REPORTING PER WORKING GROUP

WG Cargo

Status: Proceeding according to WG Plan

Within the WG Cargo major progress has been made with respect to the detailed design of the tractive and vehicle systems. The detailed design of all major vehicle and tractive scopes have been completed which consists of propulsion magnets, levitation magnets, guidance magnets, emergency braking system magnets, bogie structure, propulsion drives, magnet drives, battery, pressurized electronics enclosure, and suspension systems. For the tractive and vehicle only small, detailed design tasks remain. For infrastructure all remaining detailed design work and modifications to scopes of work have been completed as planned.

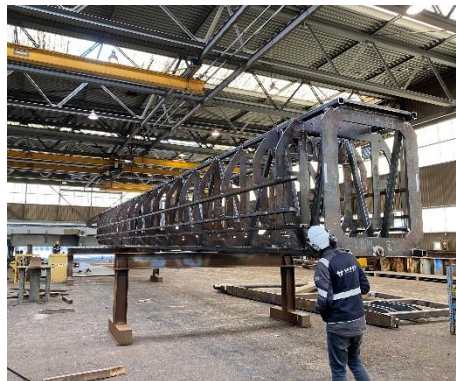
H1-2023 saw a lot of progress regarding procurement, manufacturing, and assembly of the European Hyperloop Center. Highlights include:

- Tata Steel and POSCO delivered steel plates for the switch manufacturing and Mercon has started the switch manufacturing. The first set of empty switch pipes is nearing completion.
- Many components required for the assembly of the linear and switch pipes have been ordered such as attachment structures, brackets, and safety tracks.
- Mercon has fully assembled 21 linear pipe sections and has now started switch assembly.
- The prototype of the on-board propulsion system was tested up to 300 km/h in March.

Below is a summary in images:



Welded & bend switch plates



Assembly jig for pipe sections



Flange machining



Assembled linear pipe



Linear pipe assembly



Finished linear pipes



Switch pipe manufacturing



Airlock pipe assembled



Assembled propulsion drive



Assembled and painted pipes

WG Passengers

Status: Significant delays as a result of lack of funding.

Almost no progress was made towards full-scale hyperloop passenger design due to the focus on the European Hyperloop Center and WG Cargo.

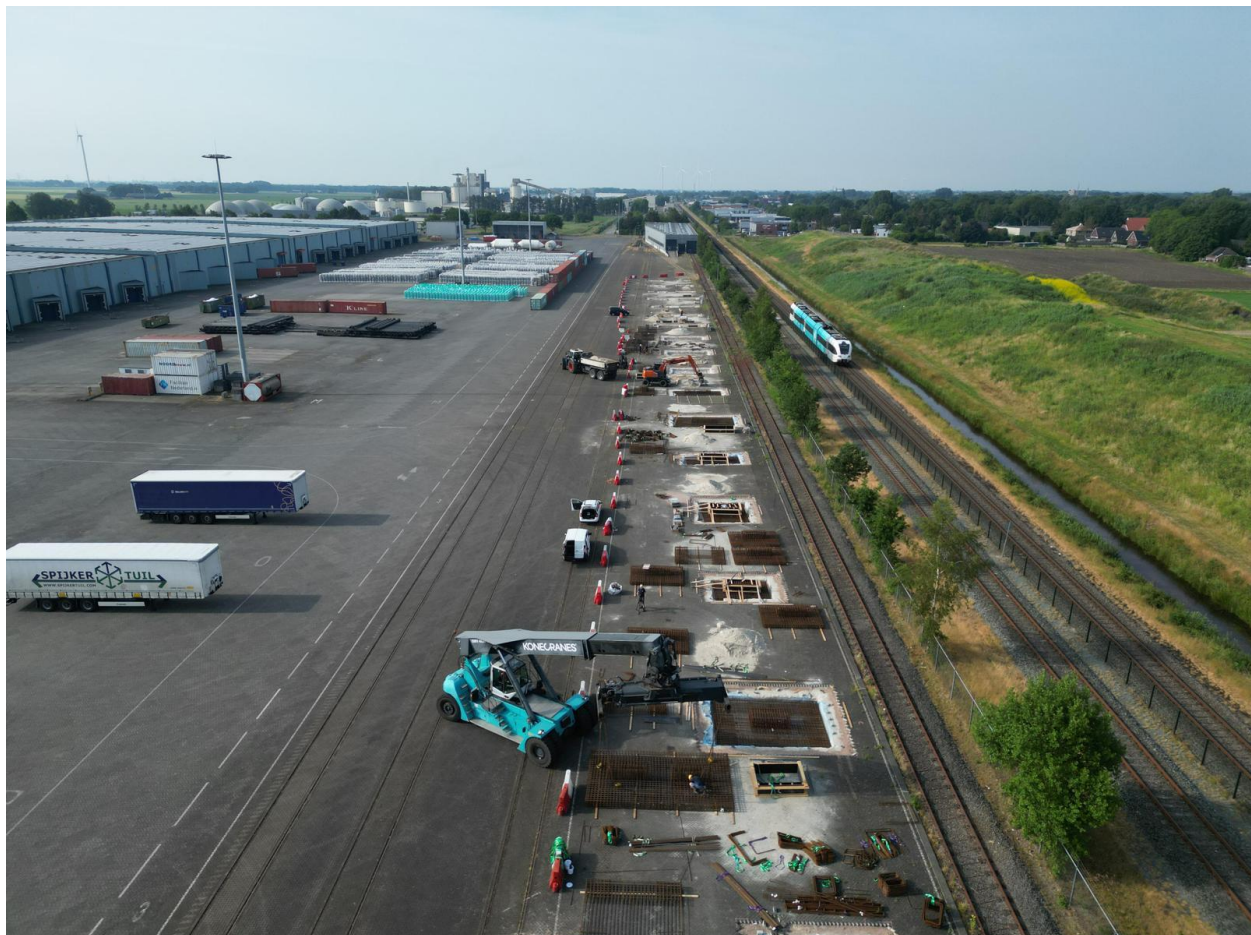
Progress was made towards standardization and regulation of the passenger hyperloop. During H1-2023 progress was made regarding hyperloop vocabulary definitions, which are almost ready to be submitted for voting. Within JTC20 new subjects and work items are currently being discussed which will be drafted in 2024 and onwards.

In H1-2023, Ramboll started their work for the European Commission on the impact assesment for hyperloop regulation.

WG EHC

Status: On-track to finish construction before end of 2023

During H1 2023 the construction of the substructure has been completed such that Hardt can install the hyperloop pipes on location in Veendam in H2 2023. Furthermore, a notification for construction and operation of the EHC has been submitted to the ODG. Preparations have been made for the realisation of the building and powersupply of the EHC in H2 2023. Expected is that the construction of the hyperloop guideway, substructure, powersupply, buildings will be finished this year.





WG Future Prospects

Status: Ongoing

Description

Three Preliminary Route studies have been done:

- Preliminary Route #3 (Delhi – Ahmedabad)
 - A study was performed to assess a hyperloop between Delhi and Ahmedabad. Hyperloop was compared to using high-speed rail. The route was selected through a strategic analysis of corridors in India. The study considers corridor characteristics, demand forecasting, journey time comparisons, modal shifts, cost estimates, and economic assessments. The study recommends to consider hyperloop as an alternative to high-speed rail for future high-speed corridors across India.
- Preliminary Route #4 (Italy)
 - In collaboration with RFI in Italy a study has been performed on the potential of a hyperloop network in Italy. RFI is the national rail administrator in Italy. The study covers route alignment, ridership forecasting, cost estimation, and cost-benefit assessment. The results are promising, showing that hyperloop could increase the overall share of rail in Italy, and has a higher than unity benefit to cost ratio.
- Preliminary Route #5 (The Netherlands)
 - A spatial integration and network study has been performed in the Netherlands, through a series of workshops organized in collaboration with the Ministry of Infrastructure and Water Management and various stakeholders. Designs have been made of how to integrate hyperloop in an urban environment, and a network has been designed that connects the provinces in the Netherlands.

EU roadmap:

- Hypervisory, an independent hyperloop advisory expert has been selected to deliver the EU Roadmap deliverable. The deliverable comprises a presentation deck and a two-pager describing the steps needed to be taken by various European stakeholders to bring hyperloop to realization. The roadmap was presented to EU stakeholders through a Hyperconnected Europe webinar.

